

VTQA2023: ACM Multimedia 2023 Visual Text Question Answering Challenge

Kang Chen
chenkangcs@stu.hit.edu.cn
Harbin Institute of Technology
Harbin, China

Tianli Zhao
tlzhaoipir@stu.hit.edu.cn
Harbin Institute of Technology
Harbin, China

Xiangqian Wu
xqwu@hit.edu.cn
Harbin Institute of Technology
Harbin, China

Image:



Chinese Annotation:

公司举行了一场饭桌会议，顾名思义就是一边吃饭一边商讨事宜。爱莲娜的金色头发果然是人群中最显眼的一个，马尔蒂格穿着橘黄色开衫，正在埋头切一块牛排；塞特温穿着深蓝色衬衫，胸前挂着会议牌，他的左手正抬起来准备去拿酒杯。塞特温的左手边，卡维利耶和阿古姆正严肃地讨论下个月的预算案。卡维利耶一头褐色齐肩短发，干净利落。

Q1: 爱莲娜穿着什么类型的上衣?

A1: 西装外套

Q2: 马尔蒂格的右手边是谁?

A2: 卡维利耶

English Translation:

The company held a table meeting, which, as the name implies, is to discuss matters while eating. **Elena's** blond hair is indeed the most conspicuous one in the crowd, and **Urtigera**, wearing an orange cardigan, is burying her head in cutting a steak; **Setwin** was wearing a dark blue shirt, with a meeting card hanging on his chest, and his left hand was lifting up to get a glass of wine. On the left of **Setwin**, **Kavelier** and **Agum** are seriously discussing the budget for next month. **Kavelier** has short brown shoulder-length hair, clean and clean.

Q1: What type of blouse does **Elena** wear?

A1: Suit

Q2: Who is on the right-hand side of **Urtigera**?

A2: **Kavelier**

Figure 1: Example in our dataset with the question-answer pairs and their corresponding image and text. Different representations of the same object in text and image are identified with the same color. For example, ‘Elena’ in the text and the object bounding box corresponding to ‘Elena’ in the image are marked red.

ABSTRACT

The ideal form of Visual Question Answering requires understanding, grounding and reasoning in the joint space of vision and language and serves as a proxy for the AI task of scene understanding. However, most existing VQA benchmarks are limited to just picking the answer from a pre-defined set of options and lack attention to text. We present a new challenge with a dataset that contains 23,781 questions based on 10124 image-text pairs. Specifically, the task requires the model to align multimedia representations of the same entity to implement multi-hop reasoning between image and text and finally use natural language to answer the question. The aim of this challenge is to develop and benchmark models that are capable of multimedia entity alignment, multi-step reasoning and open-ended answer generation.

CCS CONCEPTS

• Information systems → Question answering; • Computing methodologies → Artificial intelligence.

KEYWORDS

dataset, multimodal, visual question answering

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1 INTRODUCTION

A long-term goal of AI systems is to understand the complex real world like human beings, and Question Answering (QA) is a good task to evaluate the understanding ability of the AI system. To answer questions, people need to extract information from multiple modalities, such as text, images and structured data like knowledge base, graphs and tables. And furthermore, people need to align the information and do multi-steps reasoning between different modalities.

Although Visual Question Answering (VQA) [4] has been widely researched as a multimedia QA task, VQA models only extract information from image when answering questions and focus mainly on scene recognition, counting, color and other visual detection tasks, which do not require much logical reasoning or assignment between different modalities. Only recently, there are some attempts to introduce more modal information into VQA tasks. For example: (1) FVQA [13] and OK-VQA [8] combine knowledge base (KB) with VQA task, which requires the ability of knowledge understanding and multi-step reasoning. But it is difficult to construct a comprehensive KB in real world, which makes it restricted to answer the open-ended questions; (2) TextbookQA [5] and ScienceQA [7] use textbook as data sources, involving texts, images, tables and other multimodal information. But most images in these datasets are manually drawn schematic diagrams, and the questions are all in a multiple-choice setting, which are far from the real world.

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To address these issues, more recently, there appears some new datasets such as MultiModalQA [11] and MuMuQA [10], which involve reasoning across texts, images and so on. However, these datasets are all conducted with extracted QA. For MultiModalQA dataset, each image is associated with a Wikipedia entity, therefore the reasoning of images essentially degenerates to ranking the images and using the entities corresponding to the top-ranked images, which weakens the need for cross-media reasoning ability. As for MuMuQA, although it requires grounding between image and text along with multi-hop reasoning, there are still some problems: (1) data are all from news, which means most questions are related to human beings; (2) questions all follow a specific pattern; (3) there is only 1384 human-curated examples and the training data is generated automatically, which leads to poor quality and difficult training. Current multimedia benchmarks are still far from the real-world QA scene and cannot measure the multimedia understanding ability of AI system well.

To bridge this gap, we introduce a new challenge named “Visual Text Question Answering” (VTQA) along with a corresponding dataset, which includes only questions that require multi-hop reasoning through both image and text. The images are real images from the MSCOCO dataset [6], containing a variety of entities. Information diversity, multimedia multi-step reasoning and open-ended answer make our task more challenging than the existing datasets. All the annotations in VTQA dataset are first marked in Chinese, and we also provide the corresponding English translations.

2 DATASET

2.1 Dataset Construction

To promote the progress of this open-ended multimedia multi-hop VTQA challenge, we collect a new dataset. Our dataset consists of 10124 image-text pairs and 23,781 questions. We collect data through an annotation interface developed by ourselves. Images from MSCOCO dataset are shown in the interface and annotators are first asked to write a paragraph of text related to the image. The annotators then come up with questions based on the image-text pair and the annotation process requires that the questions cannot be answered only by image or text. Finally, the other annotators mark the answers to the questions and indicate the categories of the answers. We set three categories for answers: (1) ‘YN’ means yes-or-no answer, (2) ‘E’ means that the answer is extracted from the text, and (3) ‘G’ means that the answer is generated based on the text-image pair. And the annotators need to label ‘yes’ or ‘no’ in English for the yes-or-no answer, since there are too many words to express ‘yes’ or ‘no’ in Chinese, for example, ‘可以’ and ‘是的’ both mean ‘yes’. We promise the richness of text information by limiting the minimum length of text (more than 100 Chinese characters) and remove the questions that can be answered only by image or text to ensure the complexity of questions.

2.2 Dataset Split

We randomly split the dataset into training, validation, test-dev and test splits and each image will only appear in one split. Each split has 11312, 1245, 2189, 9035 samples, respectively. The training, validation, test-dev and test splits are provided as JavaScript Object Notation (JSON) formatted text files called train.json, val.json,

test_dev.json and test.json, where each data sample is stored as a dictionary. The attributes in the JSON file are as follows.

- `qid`: Unique id for each question.
- `image_local_path`: Source path for the image in the dataset
- `text`: Annotated text based on the image.
- `question`: Annotated question according to the image-text pair.
- `answer`: Annotated open-ended answer, usually a short phrase.
- `answer_type`: Answer type, which is described in Section 2.1.
- `yes_or_no`: Only work for the yes-or-no answers, where ‘yes’ for the positive answer and ‘no’ for negative answer. Will not appear in other types of answers.

2.3 Dataset Analysis

Images in this dataset are from the MSCOCO dataset, which contains multiple objects and rich contextual information. Fig. 2 presents the top-10 categories distribution of the images used in this dataset. Unlike MuMuQA, in which most images are related to people, images in our dataset contain more kinds of objects.

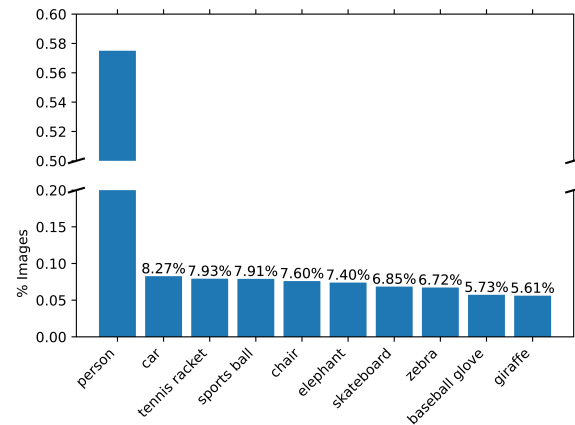


Figure 2: Top-10 categories distribution of the images used in our dataset

The texts and questions length statistics are shown in Fig. 3. We see that most texts range from 300 to 600 and most questions range from 9 to 21. Obviously, compared with the previous VQA datasets (usually less than 20 words), our dataset also puts forward higher requirements for text understanding to extract information from the long text.

3 CHALLENGE

3.1 Challenge Description

The aim of this challenge is to develop and benchmark models that are capable of multimedia entity alignment, multi-step reasoning and open-ended answer generation.

As illustrated in Fig. 1, given an image-text pair and a question, a system is required to answer the question by natural language.

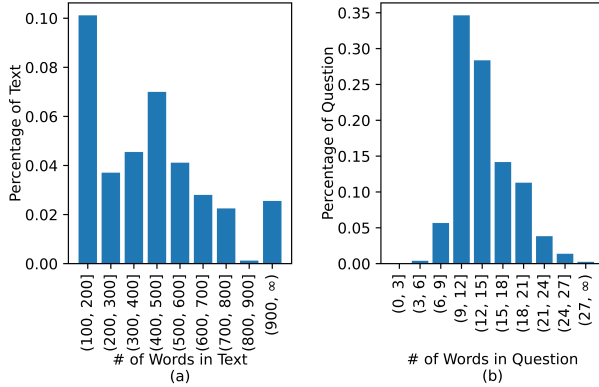


Figure 3: Percentage of questions and texts with different word lengths

Importantly, the system needs to: (1) learn to identify entities in image and text referred to the question, (2) align multimedia representations of the same entity, and (3) conduct multi-steps reasoning between text and image and output open-ended answer. For example, in Fig. 1, the key entity of Q1 is “Elena”. According to the text “gold hair”, we can determine that the second person from the right in the image is “Elena”. Finally, we further answer “suit” based on the image information. As for Q2, which is a more complex question, the previous steps need to be repeated several times to answer it.

In this challenge, we divide the English and Chinese tracks and rank them respectively.

3.2 Challenge Metric

As the answers divided into three types, we use different metrics for distinct types of answers.

Exact match (EM). This metric measures the percentage of predictions that match the ground truth answer exactly and will be used in all types of answers.

(Macro-averaged) F1 score (F1). This metric measures the average overlap between the prediction and ground truth answer. We treat the prediction and ground truth as bags of tokens and compute their F1. This metric will be used for the ‘E’ and ‘G’ types of answers.

YN accuracy (YNacc). This metric is only used for the ‘YN’ type of answers. The answer will be transformed into ‘yes’ or ‘no’ by a pre-defined yes-or-no dictionary. Then we calculate the accuracy just based on the yes-or-no answer.

The EM metric is used for ranking models and the other metrics are presented to show the performance of the algorithm and are not used for ranking.

All training and verification data are published online for participants to train their models. There are two data split for participants to eval their models: (1) Test-dev set, which is published without corresponding answers. Participants can predict answers of the test-dev set, generate and submit the JSON files to quickly evaluate

their own models. (2) Test set, which is private, and the final ranking of this challenge are based on the overall EM score of the test set. To evaluate models in test set, participants need to use a Docker image as submission. We have provided baseline implementations to minimize the overhead for participants. We evaluate the models in test split and rank the participant models by the overall EM metric.

4 EXPERIMENTS & RESULTS

4.1 Methods

Baseline[2]: We provided the baseline method and results for this challenge. The method based on multi-layer key entity extraction (KEE) and cross-media reasoning (CMR) on the extracted entities, named KECMR. The KEE module extracts the main entities in text and image according to the relationship between the features of question, text and image. The CMR module perform entity-level and media-level attention-based reasoning among the key entities extracted by KEE module, and then put the features back to their original positions, which blocks the interactions with unimportant entities. The two modules are stacked in multiple layers, and it allows the model to select different entities in each layer. This structure provides flexibility and controllability in training, and reduces the calculation complexity.

MGN[15]: Zhang et al. proposed a method using a multimodal encoder for reasoning and an autoregressive decoder for open-ended answer generation. They argue that the main challenge of the VTQA task is to align entities across different modalities and open-ended answers. First, it encodes the image through the ViT[3]. Then it cascades the image feature with text and question embeddings and feeds them into a Transformer[12] for multi-modal reasoning, which allows the reasoning process to focus on information completion. Finally, a Transformer Decoder generate open-ended answer autoregressively according to the encoder outputs. The multi-modal Transformer encoder and decoder are based on T5[9] and fine-tune in VTQA dataset.

VTQAGen[1]: Chen et al. proposed a generative method based on multimodal attention alignment and pre-trained language models. Similar to MGN, they consider that the challenge of the VTQA task comes from multimodal entity alignment and open-ended answer generation, yet apply different pre-trained models and different architectures. First, the method cascades image region features generated by Faster R-CNN with text and questions. Then it performs multimodal reasoning based on attention and the aligned features are then fed entirely to BART, a language model, to generate coherent open-ended answers. In the autoregressive decoding stage, it applied a token-based multi-model ensemble approach to generate more reliable open-ended answers.

AEEA[14]: Yu et al. proposed an improved method based on baseline (KECMR). They improved the KEE module to Aligned Key Entity Extraction (AKEE). It aligns entities in text and image by adding a multi-layer cross-modal attention (MCA) module before the original modality fusion module, which prevent wrong entity extraction caused by semantic misalignment of image and text features. It also improves CMR module to unified attention cross-modal reasoning (UCMR) by merging the original two stage attention. It

Table 1: Top 3 results of the VTQA Challenge in test set

Method	EM	YN-Acc	E-F1	G-F1
English Track				
MGN	66.38	80.55	81.82	46.81
VTQAGen	59.46	75.24	73.27	45.34
AEEA	58.44	75.06	69.82	41.71
Baseline	56.32	73.01	66.58	43.73
Chinese Track				
MGN	61.27	75.18	79.57	55.10
VTQAGen	58.59	75.94	74.63	51.57
AEEA	-	-	-	-
Baseline	51.31	77.63	60.47	51.12

use a parameter matrix to explicitly regulate the intra-modal attention and cross-modal attention. Finally, for non-"Yes/No" type questions, it adds an attention map guided answer injection auxiliary loss, which reduces the difference of attention map on image or text between question and answer. It teaches the model that the question and answer should concentrate on the same area in image or text.

4.2 Results & Analysis

A total of 46 participants have registered on the website for this challenge. They are mainly from colleges, universities, enterprises and scientific research institutes in China, India and United States. This challenge is divided into two tracks, Chinese and English. For the Chinese track, there are 29 effective submissions on the test-dev set, and 6 effective submissions on the test set. For the English track, there are 105 effective submissions on the test-dev set and 8 effective submissions on the test set.

The top two teams are generative methods based on pre-trained models, and the third is a classification method based on improved baselines. The result comparison with baseline on the test set is shown in Table 1. MGN and VTQAGen uses the pre-trained language model to perform better multi-modal entity alignment and open-ended answer generation. AEEA improves the performance of the baseline method by enhancing the interaction between modalities before entity extraction, controlling the interaction area, and the loss of answer semantic alignment.

The generative methods MGN and VTQAGen are based on pre-trained model fine-tuning. MGN using ViT and T5 has a very large advantage, which shows that better image and text features are beneficial for VTQA task, which is directly related to whether the model can perform correct information transfer between question, image and text through semantic similarity and find the correct answer. Meanwhile, it could be figure out that VTQAGen and AEEA have comparable performance on "Yes/No" type questions, while AEEA lags behind VTQAGen on Extract and Generate type questions, which proves that the generative answer is indeed more in line with the open-ended form of VTQA. It is probably because using an answer vocabulary for classification would lead to a severe class imbalance, which in turn would cause rare classes to be ignored.

On the Chinese track, methods based on pre-trained models show a significant advantage over the baseline, only except on "Yes/No" type questions. In Chinese, "Yes/No" could have multiple ways to express according to the context, such as "是/否", "有/没有", "可以/不能", "在/不在", etc. The model not only needs to determine whether the fact described by the question is correct, but also needs to find out which type the fact belongs to, which requires more complex and more accurate reasoning during answer generation. We speculate that simple fine-tuning method may not be able to learn this ability.

5 CONCLUSION & FUTURE CHALLENGES

In this challenge, we provide the VTQA dataset, which is completely manually labeled with text, questions and generated answers in real-world scenarios, making this task more challenging than previous similar works. During the challenge, we received the support from many researchers, and we discovered many problems, such as shortcuts and common-sense information in the dataset, which gave us the motivation on future working. In our future work, we hope to make the annotation richer, more specific and more balanced. Moreover, we would like to make the reasoning more complex and logical, even more confusing, in an attempt to evaluate the reasoning ability of pre-trained large models.

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